



Q1. How can I use ESP-12F module in place of ESP-12E? I have an ESP-12F module but am unable to upload the code to it. Please help.

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A1. ESP series modules are Wi-Fi modules and are popular in IoT projects. However, there are some confusions between ESP modules due to lack of information on the modules. One of the main differences between ESP-12E and ESP-12F modules is the antenna design. Otherwise, both have same footprints.

You can replace ESP-12F with ESP-12E in most cases without problem. You can use both the boards as NodeMCU 1.0 (ESP-12E module) in Arduino IDE. But if you have a bare ESP-12F module then you need some manual setup as explained here.

On the Internet, you will come across many complaints, such as not working or error while uploading the code to ESP-12F module. The problems could be one or many. First, the most common one is the power supply.

ESP-12F module works on 3.3V DC supply but some users accidentally connect it to 5V supply, which can damage the module. To use it with other microcontrollers and Arduino boards, it is recommended to use a 3.3V to 5V level shifter module.

If you are using it with USB-to-TTL adaptor for connecting to PC, you need some set up like adding pull-up resistor (10k) to pin GPIO0 and pull-down resistor to pin GPIO15. Depending on the board, you also need to pull up RST and DTR pins. You need to download the appropriate USB driver and install it in your system prior to connecting it to your PC.

However, if you have NodeMCU version, things are simple. The NodeMCU boards are pre-wired with ESP-12F module and other associated components, so all the relevant GPIO pin functions can be performed by the Arduino IDE.

Typical ESP-12F module and NodeMCU version board are shown in Figs 1 and 2, respectively.



Fig. 1: ESP-12F module



Fig. 2: NodeMCU ESP-12F module

Q2. How do you determine the resistor values in a circuit? Please explain the selection process in brief.

Abin

A2. Resistor values are selected after analysis for a given application. For most of the regular circuits, 1/4W resistors with $\pm 5\%$ tolerance carbon composition or carbon film resistors are used. However, for power applications, wire-wound, ceramic, and metal alloy resistors are commonly used.

A resistor of at least twice the calculated power dissipation is selected. If resistor is for power application, a resistor of at least three or four times higher than the calculated power dissipation value should be selected.



Fig. 3: Different types of fixed resistors

For commercial circuits or professionally designed circuits, the designer has to consider many other parameters as well while selecting a resistor for the circuit. It may involve scrutinising and analysing the datasheets provided by the manufacturer of the resistors.

There are several ranges of resistors available for electronics industry. The International Electrotechnical Commission is an international standards organisation that prepares and publishes international standards for all electrical, electronic, and related products including resistors. The standard range of resistors are grouped under E series: E3, E6, E12, E24, E48, E96, and E192.

For example, the standard resistances under E12 series are

10 Ω , 12 Ω , 15 Ω , 18 Ω , 22 Ω , 27 Ω , 33 Ω , 39 Ω , ... till 10M Ω . The values in the range are chosen such that they follow an exponential curve, and when you replace your calculated value with one from the preferred range, the error will be within a fixed percentage. Some resistors are colour coded to indicate the values or ratings while some resistors have the values printed with numbers or letters on the bodies.

Different types of fixed resistors commonly used in the circuits are shown in Fig. 3. The most common process of selecting a resistor for a given application involves the following steps:

1. Selection of a resistor through a quick circuit analysis.
2. Checking the parameters like nominal value, tolerance, power rating, voltage drop, current through it, power dissipation by it and other properties, if required.
3. Calculating the power requirement of the desired resistor. A resistor of at least twice the calculated power dissipation should be selected. If it is a power application, the resistor of at least three or even four times the calculated power dissipation would be better.
4. Checking if the exact value of that resistor is available. If not, use parallel or series combination of resistors to get the required resistance value.
5. Deciding the type of resistor suitable for the given application or circuit. For example, through-hole or surface mount device (SMD), trimmable, variable, fixed type, etc.